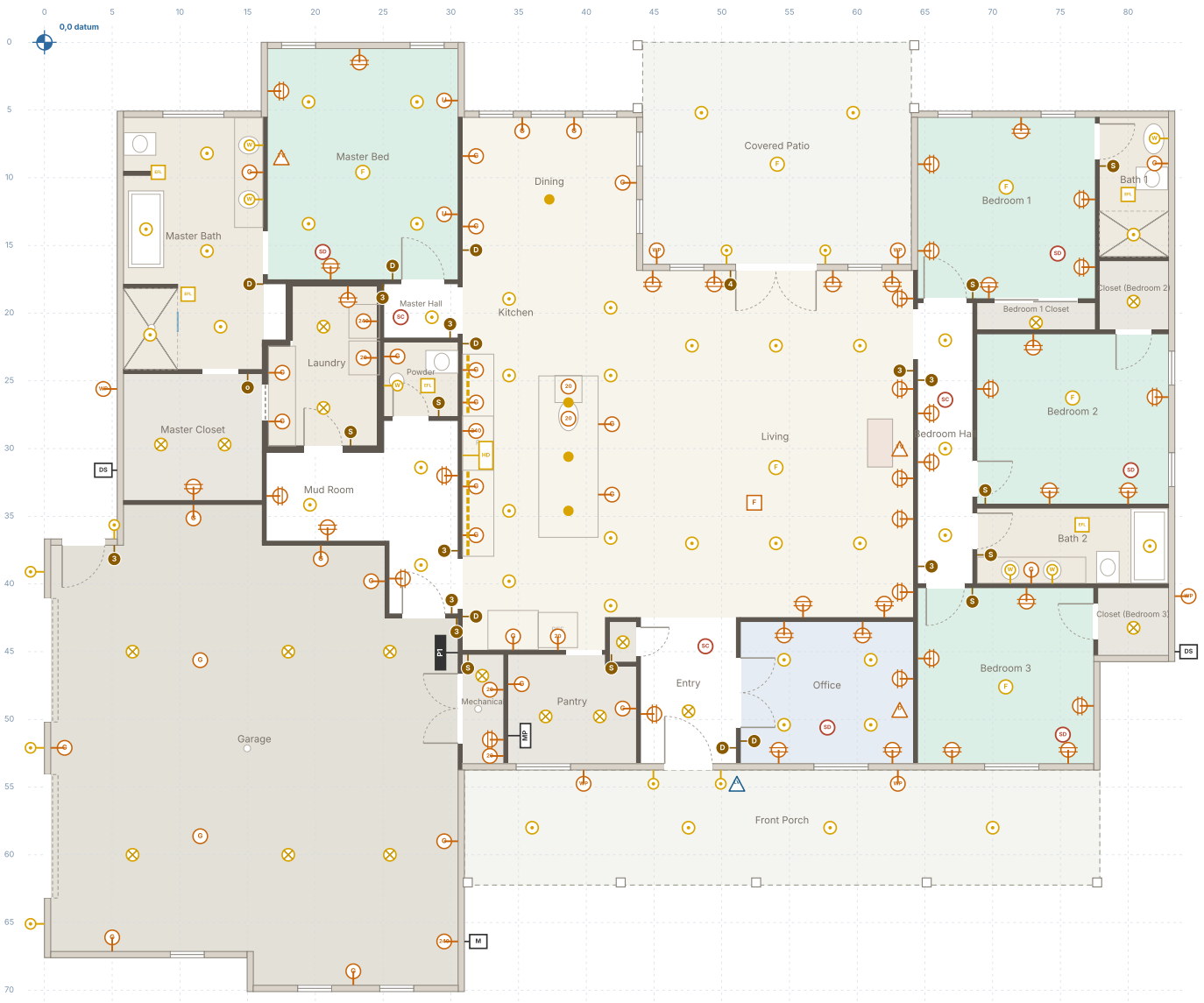
REAR · PATIO



FRONT · PORCH

duplex receptacle · GFCI · duplex + USB-C · 20A dedicated · 240V · weatherproof GFCI · TV box / Cat6

switch S / 3 / 4 / D dimmer / o occupancy · recessed LED · surface / closet LED · pendant · fan-rated box · exhaust fan (+ light)

vanity light / sconce · exterior light · smoke SD / smoke+CO SC · panel P1 · disconnect DS / meter M

Receptacles R, switches S, lights and fans L, alarms A, panels and equipment X. · Pick a device for its circuit and switch legs; pick a circuit tab to see one circuit and its run from P1. · Panel P1 in the garage. Calculated load 150 A on a 200 A service.

draft

E-1 Electrical · 64 Stone Sheep Circle

Devices, switch legs and circuits for panel P1, drawn on the walls they go in. App version v5. From the owner's plans (Larson House, 6/11/26).

Notes

1. Panel P1, 200A main breaker, 40 spaces, in the garage on the mud room wall beside D6, 60" to centre, 36" clear in front (110.26). It sits next to the mechanical room (boiler, water heater) and the kitchen, the two heaviest groups of circuits. A meter-main outside on the garage's porch-side wall carries the service disconnect (NEC 2023 230.85 emergency disconnect) and feeds P1 with 4-wire 2/0 Al SER (or #2/0 Al in conduit), neutral isolated in P1.
2. Code basis: NEC 2023 (confirm the edition Powell / Park County enforces). AFCI on every 120V 15A and 20A branch circuit in the house except the bath receptacle circuits and the garage and exterior circuits (210.12); GFCI in baths, kitchen counters and island, dishwasher, disposal, laundry, garage, outdoors, and within 6 ft of any sink (210.8). Tamper-resistant receptacles throughout (406.12).
3. Positions: every wall device is drawn on the face of the wall it goes in, with its stem to that wall; the schedule gives the wall and the distance from the nearer corner. Ceiling devices give two distances. Final boxes are set to the studs at the owner walk-through.
4. Mounting heights (standard): receptacles 16" to centre, switches 48", counter and laundry GFCIs 44", vanity GFCIs 40", vanity lights 84", exterior lights 72" or 84".
5. Switch plan: every light belongs to one switch leg (table below), switched in the room it lights, at the latch side of the door you walk in through. Rooms with two ways in get 3-ways: the master hall, mud room, kitchen, bedroom hall and garage; the living cans have three control points (entry dimmer, 4-way at the patio doors, 3-way at the bedroom hall). A switch tag that appears in several legs is a multi-gang bank, one rocker per leg. On the page, picking a light or switch lights every device on its legs in blue.
6. Circuit sheets E-1.1 onward show one circuit each with its cable run drawn from P1 through every device in nearest-neighbour order (solid for receptacle and appliance circuits, dashed for lighting). The run is a proposed order, not a required one. E-1.D shows the Cat6 home runs to the media panel.

Open items

- Boiler and water heater fuel: gas or propane is drawn (200A service). An electric boiler or air-to-water heat pump with an electric water heater adds about 80-120A and needs a 320/400A service.
- Range fuel: the 48" range is drawn electric on a 50A circuit. A gas or dual-fuel range drops that to a 120V receptacle and adds a gas line.
- Mini-split breakers are sized from M-1's nominal units (CU-1 18k on 20A/#12, CU-2 36k on 40A/#8; 30A/#10 if the chosen model's MOP allows); set them to the nameplate MCA/MOCP of the units bought. If the living head moves to CU-1, CU-1 grows and 5-6 goes to 25-30A.
- Utility service: overhead or underground, and which side of the house the meter goes on. The meter-main is drawn on the garage wall facing the porch end.
- EV-ready: a 50A NEMA 14-50 receptacle is drawn in the garage. Keep, swap for a hardwired charger, or drop?
- Mounting heights: standard heights are drawn. Zero Rd used 30" receptacles and 42" switches for accessibility; say if this house should match.
- Ceiling fans in the master, living, three bedrooms and the patio: keep the fan-rated boxes, or cans instead?
- Low voltage: TVs in the bedrooms, ceiling Wi-Fi access points, cameras, and whether the media panel should move out of the pantry.

7. Kitchen: three 20A small-appliance circuits (11 left counter, 12 island and front counter, 13 dining and pantry, 210.11(C)(1)), counter receptacles at 4 ft spacing (210.52(C)); range, refrigerator, dishwasher, disposal and hood each on their own circuit. The island has a receptacle in each end panel on the living side, below the counter edge. Kitchen and living ceilings are vaulted 4:12: sloped-ceiling can trims and a fan downrod sized for the slope.
8. Lighting circuits by area, whole switch legs only: 33 master suite, master hall, laundry, powder, mud room; 34 kitchen, dining, pantry, mech room, living, entry, coat closet, porch and patio; 35 office, bedroom wing and every smoke and CO alarm (so a tripped alarm circuit shows as dark lights). Garage and garage exterior lights on 36. Each is well under the 1,440 W continuous limit of a 15A circuit at LED design wattages.
9. Smoke alarms in every bedroom and the office; smoke + CO combination alarms outside each sleeping area (master hall, bedroom hall, entry outside the office). All 120V, interconnected, battery backup (IRC R314, R315; CO because of the attached garage, the wood fireplace and the gas boiler).
10. Bath exhaust fans 110 CFM with humidity sensing and an integrated LED light: the light switches with the room's other lights, the fan on its own rocker with a timer (14/3 from the bank). Ducted outside in rigid duct.
11. Mini-splits: each outdoor unit has its own 2-pole circuit and a fused disconnect within sight; the indoor heads are fed from their outdoor unit through the interconnect cable, so they have no circuits of their own. CU-1 (18k) sits in the left side yard on the master closet wall and CU-2 (36k) in the right side yard behind the bedroom 3 closet, per M-1; breakers are sized from the nominal units until the nameplates are known. A weatherproof GFCI receptacle is within 25 ft of each outdoor unit (210.63).
12. Radiant (R-1): the wall-hung boiler HS-1, its system circulator and the zone controller share 15A circuit 31 with a service switch at the boiler, and each remote manifold station has a receptacle on 31 for its actuator wiring module (MF-1 laundry, MF-2/MF-G on the garage side of the mud room wall with the garage circulator, MF-3 bedroom hall). The water heater's power vent is on 32. Thermostats are 24V back to the zone controller, not line voltage. The MF-3 cabinet sits below the hall 3-way S at 48".

13. Low voltage: Cat6 home runs to the media panel on the pantry's mech-room wall (master and living TV boxes, office desk, video doorbell at the front door), powered from circuit 37.

Circuit schedule, panel P1

Ckt	Load	Breaker	Wire	Serves	Devices
1-2	Range, 48" electric	240V 50A 2P	#6 Cu	kitchen, left wall	1
3-4	Dryer	240V 30A 2P	#10 Cu	laundry	1
5-6	Mini-split CU-1, 18k (master, dining heads)	240V 20A 2P	#12 Cu	left side yard, master closet wall; heads fed from the unit (MCA ~17A)	1
7-8	Mini-split CU-2, 36k (bedrooms 1-3, living heads)	240V 40A 2P	#8 Cu	right side yard, bedroom 3 closet wall; heads fed from the unit (MCA ~28A, MOP 35-40A)	1
9-10	EV-ready receptacle (NEMA 14-50)	240V 50A 2P	#6 Cu	garage, front wall	1
11	Kitchen small appliance 1	20A GFCI/AFCI	#12	left counter run	4
12	Kitchen small appliance 2	20A GFCI/AFCI	#12	island, front counter	3
13	Small appliance 3: dining + pantry	20A GFCI/AFCI	#12	dining walls, pantry counter	7
14	Refrigerator	20A AFCI	#12	kitchen, front wall	1
15	Dishwasher	20A GFCI/AFCI	#12	island	1
16	Disposal	20A GFCI/AFCI	#12	island sink base	1
17	Range hood	15A AFCI	#14	over the range	1
18	Laundry: washer + counter	20A GFCI/AFCI	#12	laundry	3
19	Bath receptacles west	20A GFCI	#12	master bath, powder	2
20	Bath receptacles east	20A GFCI	#12	bath 1, bath 2	2
21	Master suite receptacles	15A AFCI	#14	master bedroom, master closet	7
22	Living receptacles	15A AFCI	#14	living, TV wall, floor box, fireplace blower	13
23	Entry + office receptacles	15A AFCI	#14	entry, office	6
24	Bedroom 1 + hall receptacles	15A AFCI	#14	bedroom 1, bedroom hall	7
25	Bedroom 2 receptacles	15A AFCI	#14	bedroom 2	5
26	Bedroom 3 receptacles	15A AFCI	#14	bedroom 3	5

Ckt	Load	Breaker	Wire	Serves	Devices
27	Mud room receptacles	20A GFCI/AFCI	#12	mud room	4
28	Garage receptacles	20A GFCI	#12	garage walls, one per bay	6
29	Garage door openers	20A GFCI	#12	two ceiling receptacles	2
30	Exterior receptacles	20A GFCI	#12	porch, patio, both mini-split units	6
31	Boiler, circulators, zone controls	15A AFCI	#14	mechanical room, the three manifold stations (MF-1 laundry, MF-2/MF-G garage, MF-3 hall)	5
32	Water heater (power vent)	15A AFCI	#14	mechanical room	1
33	Lighting A	15A AFCI	#14	master suite, master hall, laundry, powder, mud room (~419 W LED at design wattages)	33
34	Lighting B	15A AFCI	#14	kitchen, dining, pantry, mech, living, entry, coat closet, porch, patio (~564 W LED at design wattages)	45
35	Lighting C + alarms	15A AFCI	#14	office, bedroom wing, baths, closets, every smoke and CO alarm (~490 W LED at design wattages)	36
36	Garage + garage exterior lights	15A	#14	garage, garage door and service door lights (~156 W LED at design wattages)	12
37	Media panel + low voltage	15A AFCI	#14	media panel in the pantry, doorbell transformer	3
38-40	Spare	—	—	three spaces	0

Switch plan

One row per switch leg. Pick a row to light up its switch and lights on the drawing.

Leg	What	Room	Switch(es)	Lights	Control	Notes
mbed-cans	Master bedroom cans	Master Bed	S1 dimmer	L1, L2, L3, L4	dimmer at the bedroom door, rocker 1 of 3	
mbed-fan	Master bedroom fan box	Master Bed	S1 dimmer	L5	rocker 2 at the bedroom door; fan speed on the fan remote	fan-rated box for a ceiling fan with light kit
mhall	Master hall	Master Hall	S1 dimmer; S2 3-way switch; S3 3-way switch	L6	3-way: kitchen opening and the bedroom door (rocker 3 of the bedroom bank is its partner on the hall side)	
mbath-light	Master bath lights	Master Bath	S4 dimmer	L7, L8, L9, L10, L11, L14, L15	dimmer at the bath opening, rocker 1 of 3	cans, vanity lights and the fan units' lights dim together
mbath-fan	Master bath fans	Master Bath	S4 dimmer	L14, L15	rocker 2 with a 60-minute timer; humidity sensing runs them anyway	
mbath-shower	Master shower and tub cans	Master Bath	S4 dimmer	L12, L13	rocker 3 at the bath opening	wet-rated

Leg	What	Room	Switch(es)	Lights	Control	Notes
mcloset	Master closet	Master Closet	S5 occupancy-sensor switch	L16, L17	occupancy-sensor switch at the bath opening	closet-rated fixtures, 12" from shelves (410.16)
laundry	Laundry	Laundry	S6 switch	L18, L19	switch at the laundry door, latch side	
powder-light	Powder lights	Powder	S7 switch	L20, L21	rocker 1 at the powder door	vanity light and the fan unit's light
powder-fan	Powder fan	Powder	S7 switch	L21	rocker 2 with a 20-minute timer	
mud	Mud room cans	Mud Room	S8 3-way switch; S9 3-way switch	L22, L23, L24	3-way: garage door and the kitchen opening	
kit-cans	Kitchen cans	Kitchen	S10 dimmer; S11 dimmer	L26, L27, L28, L29, L30, L31, L32, L33	3-way dimmer: mud room opening and the master hall opening	sloped-ceiling trims under the 4:12 vault
kit-pend	Island pendants	Kitchen	S10 dimmer	L34, L35, L36	dimmer at the mud room opening, rocker 2	
kit-ucl	Under-cabinet lights	Kitchen	S10 dimmer	L37, L38	rocker 3 at the mud room opening	LED tape with a driver in the upper cabinet
dining	Dining chandelier	Dining	S12 dimmer	L39	dimmer on the dining wall at the kitchen end	
pantry	Pantry	Pantry	S13 switch	L40, L41	switch inside the pantry opening	
mech	Mechanical room	Mechanical	S14 switch	L42	switch inside the door	
living-cans	Living cans	Living	S15 4-way switch; S16 3-way switch; S17 dimmer	L43, L44, L45, L46, L47, L48	3 places: dimmer at the entry, 4-way at the patio doors, 3-way at the bedroom hall	sloped-ceiling trims under the 4:12 vault
living-fan	Living fan box	Living	S15 4-way switch	L49	rocker 2 at the patio doors; fan speed on the fan remote	downrod sized for the 4:12 vault
entry	Entry light	Entry	S17 dimmer	L50	rocker at the front door	
coat	Coat closet	Coat Closet	—	L51	integral occupancy sensor, no switch	no wall space for a switch inside a 2 ft closet
porch	Porch cans and front door lights	Front Porch	S17 dimmer	L78, L79, L80, L81, L82, L83	rocker at the front door; photocell or timer optional	wet-rated, dark-sky
patio-lights	Patio cans and door lights	Covered Patio	S15 4-way switch	L84, L85, L87, L88	rocker 3 at the patio doors	wet-rated
patio-fan	Patio fan	Covered Patio	S15 4-way switch	L86	rocker 4 at the patio doors	wet-rated fan
office	Office cans	Office	S18 dimmer	L52, L53, L54, L55	dimmer at the office doors	
bhall	Bedroom hall cans	Bedroom Hall	S19 3-way switch; S20 3-way switch	L56, L57, L58	3-way: living end and bedroom 3 end	
bed1-light	Bedroom 1 light	Bedroom 1	S21 switch	L59	rocker 1 at the bedroom door (light kit on the fan)	

Leg	What	Room	Switch(es)	Lights	Control	Notes
bed1-fan	Bedroom 1 fan	Bedroom 1	S21 switch	L59	rocker 2 at the bedroom door	fan-rated box
bed1-closet	Bedroom 1 closet	Bedroom 1 Closet	—	L60	integral occupancy sensor, no switch	closet-rated, 12" from shelves (410.16)
bed2-light	Bedroom 2 light	Bedroom 2	S23 switch	L64	rocker 1 at the bedroom door (light kit on the fan)	
bed2-fan	Bedroom 2 fan	Bedroom 2	S23 switch	L64	rocker 2 at the bedroom door	fan-rated box
bed2-closet	Bedroom 2 closet	Closet (Bedroom 2)	—	L65	integral occupancy sensor, no switch	closet-rated, 12" from shelves (410.16)
bed3-light	Bedroom 3 light	Bedroom 3	S25 switch	L70	rocker 1 at the bedroom door (light kit on the fan)	
bed3-fan	Bedroom 3 fan	Bedroom 3	S25 switch	L70	rocker 2 at the bedroom door	fan-rated box
bed3-closet	Bedroom 3 closet	Closet (Bedroom 3)	—	L71	integral occupancy sensor, no switch	closet-rated, 12" from shelves (410.16)
bath1-light	Bath 1 lights	Bath 1	S22 switch	L61, L62, L63	rocker 1 at the bath door	vanity light, shower can and the fan unit's light
bath1-fan	Bath 1 fan	Bath 1	S22 switch	L62	rocker 2 with a 60-minute timer	
bath2-light	Bath 2 lights	Bath 2	S24 switch	L66, L67, L68, L69	rocker 1 at the bath door	vanity lights, tub can and the fan unit's light
bath2-fan	Bath 2 fan	Bath 2	S24 switch	L68	rocker 2 with a 60-minute timer	
garage	Garage lights	Garage	S26 3-way switch; S27 3-way switch	L72, L73, L74, L75, L76, L77	3-way: service door and mud room door	
garage-ext	Garage door and service door lights	Outside	S27 3-way switch	L89, L90, L91, L92	rocker 2 at the mud room door; photocell optional	dark-sky

Device schedule

Tag	Device	Position	AFF	Circuit	Notes
R1	duplex receptacle, 15A, tamper-resistant	Master Bed: on the rear wall, 6'-9 1/2" from the left wall	16"	21	
R2	duplex receptacle with USB-C, 15A	Master Bed: on the right wall, 3'-10" from the rear wall	16"	21	nightstand
R3	duplex receptacle with USB-C, 15A	Master Bed: on the right wall, 4'-10" from the front wall	16"	21	nightstand
R4	duplex receptacle, 15A, tamper-resistant	Master Bed: on the front wall, 4'-6 1/2" from the left wall	16"	21	
R5	duplex receptacle, 15A, tamper-resistant	Master Bed: on the left wall, 3'-1 1/2" from the rear wall	16"	21	

Tag	Device	Position	AFF	Circuit	Notes
R6	recessed TV box: power + 2 x Cat6	Master Bed: on the left wall, 8'-1/2" from the rear wall	60"	21	TV facing the bed
S1	dimmer, rocker	Master Bed: on the front wall, 4'-10" from the right wall	48"	33	3-gang at the bedroom door, latch side of D8: cans (dimmer), fan box, master hall 3-way; legs mbed-cans, mbed-fan, mhall
L1	4" recessed LED, 2700K, IC, dimmable	Master Bed: ceiling, 3'-1/2" from the left wall, 3'-11 1/2" from the rear wall	ceiling	33	
L2	4" recessed LED, 2700K, IC, dimmable	Master Bed: ceiling, 3'-1/2" from the right wall, 3'-11 1/2" from the rear wall	ceiling	33	
L3	4" recessed LED, 2700K, IC, dimmable	Master Bed: ceiling, 3'-1/2" from the left wall, 4'-1 1/2" from the front wall	ceiling	33	
L4	4" recessed LED, 2700K, IC, dimmable	Master Bed: ceiling, 3'-1/2" from the right wall, 4'-1 1/2" from the front wall	ceiling	33	
L5	fan-rated ceiling box, switched	Master Bed: ceiling, 7'-1/2" from the left wall, 7'-11 1/2" from the front wall	ceiling	33	room centre, below the label
A1	smoke alarm, 120V interconnected w/ battery	Master Bed: ceiling, 4'-1 1/2" from the left wall, 1'-11 1/2" from the front wall	ceiling	35	over 3 ft from the fan blades
S2	3-way switch, rocker	Master Hall: on the rear wall, 7 1/2" from the left wall	48"	33	hall side of the bedroom door; legs mhall
S3	3-way switch, rocker	Master Hall: on the front wall, 7" from the right wall	48"	33	at the kitchen opening; legs mhall
L6	4" recessed LED, 2700K, IC, dimmable	Master Hall: ceiling, 1'-11 1/2" from the right wall, 1'-11 1/2" from the rear wall	ceiling	33	
A2	smoke + CO alarm, 120V interconnected w/ battery	Master Hall: ceiling, 1'-3" from the left wall, 1'-11 1/2" from the rear wall	ceiling	35	outside the master suite
R7	GFCI receptacle, 20A	Master Bath: on the right wall, 4'-1/2" from the rear wall	40"	19	between the two vanity sinks
L7	wall sconce / vanity light	Master Bath: on the right wall, 2'-1/2" from the rear wall	84"	33	over the mirror at sink 1
L8	wall sconce / vanity light	Master Bath: on the right wall, 6'-1/2" from the rear wall	84"	33	over the mirror at sink 2
S4	dimmer, rocker	Master Bath: on the right wall, 6'-4" from the front wall	48"	33	3-gang just inside the opening from the bedroom: lights (dimmer), fans, shower and tub cans; legs mbath-light, mbath-fan, mbath-shower
L9	4" recessed LED, 2700K, IC, dimmable	Master Bath: ceiling, 4'-2" from the right wall, 2'-8" from the rear wall	ceiling	33	
L10	4" recessed LED, 2700K, IC, dimmable	Master Bath: ceiling, 4'-2" from the right wall, 8'-11 1/2" from the front wall	ceiling	33	
L11	4" recessed LED, 2700K, IC, dimmable	Master Bath: ceiling, 3'-2" from the right wall, 3'-2" from the front wall	ceiling	33	
L12	4" recessed LED, wet location	Master Bath: ceiling, 1'-8" from the left wall, 8'-3" from the rear wall	ceiling	33	over the tub

Tag	Device	Position	AFF	Circuit	Notes
L13	4" recessed LED, wet location	Master Bath: ceiling, 1'-11 1/2" from the left wall, 2'-7" from the front wall	ceiling	33	in the shower
L14	exhaust fan with integrated LED light, 110 CFM, humidity sensing	Master Bath: ceiling, 2'-7" from the left wall, 4'-1/2" from the rear wall	ceiling	33	toilet; the light dims with the room, the fan on rocker 2
L15	exhaust fan with integrated LED light, 110 CFM, humidity sensing	Master Bath: ceiling, 4'-9" from the left wall, 5'-7" from the front wall	ceiling	33	between the tub and the shower
S5	occupancy-sensor switch	Master Closet: on the rear wall, 1'-1" from the right wall	48"	33	inside the opening from the bath; legs mcloset
L16	surface LED, closet-rated (410.16), integral occupancy sensor	Master Closet: ceiling, 2'-9" from the left wall, 4'-5 1/2" from the front wall	ceiling	33	
L17	surface LED, closet-rated (410.16), integral occupancy sensor	Master Closet: ceiling, 2'-9 1/2" from the right wall, 4'-5 1/2" from the front wall	ceiling	33	
R8	duplex receptacle, 15A, tamper-resistant	Master Closet: on the front wall, 5'-1" from the right wall	16"	21	steamer, vacuum
R9	receptacle, 20A dedicated	Laundry: on the right wall, 5'-3 1/2" from the rear wall	42"	18	washer, behind the machine
R10	240V receptacle or junction	Laundry: on the right wall, 2'-7" from the rear wall	30"	3-4	dryer, NEMA 14-30R
R11	GFCI receptacle, 20A	Laundry: on the left wall, 5'-5" from the front wall	44"	18	laundry counter
R12	GFCI receptacle, 20A	Laundry: on the left wall, 1'-9 1/2" from the front wall	44"	18	laundry counter
S6	switch, rocker	Laundry: on the front wall, 2'-0" from the right wall	48"	33	latch side of D12; legs laundry
L18	surface LED ceiling fixture	Laundry: ceiling, 4'-0" from the right wall, 3'-0" from the rear wall	ceiling	33	
L19	surface LED ceiling fixture	Laundry: ceiling, 4'-0" from the right wall, 2'-9 1/2" from the front wall	ceiling	33	
R13	GFCI receptacle, 20A	Powder: on the left wall, 1'-1" from the rear wall	40"	19	beside the lav
L20	wall sconce / vanity light	Powder: on the left wall, 2'-8" from the front wall	84"	33	over the mirror
S7	switch, rocker	Powder: on the front wall, 1'-5 1/2" from the right wall	48"	33	2-gang, latch side of D13: light, fan; legs powder-light, powder-fan
L21	exhaust fan with integrated LED light, 110 CFM, humidity sensing	Powder: ceiling, 2'-3" from the right wall, 2'-6 1/2" from the front wall	ceiling	33	
R14	duplex receptacle, 15A, tamper-resistant	Mud Room: on the left wall, 5'-7" from the rear wall	16"	27	bench
R15	duplex receptacle, 15A, tamper-resistant	Mud Room: on the front wall, 4'-6 1/2" from the left wall	16"	27	
R16	duplex receptacle, 15A, tamper-resistant	Mud Room: on the left wall, 2'-7 1/2" from the front wall	16"	27	

Tag	Device	Position	AFF	Circuit	Notes
R17	duplex receptacle, 15A, tamper-resistant	Mud Room: on the right wall, 4'-1" from the rear wall	16"	27	
S8	3-way switch, rocker	Mud Room: on the front wall, 6" from the right wall	48"	33	at the garage door, latch side of D6; legs mud
S9	3-way switch, rocker	Mud Room: on the right wall, 4'-8" from the front wall	48"	33	at the kitchen opening; legs mud
L22	4" recessed LED, 2700K, IC, dimmable	Mud Room: ceiling, 3'-2 1/2" from the left wall, 5'-8" from the rear wall	ceiling	33	
L23	4" recessed LED, 2700K, IC, dimmable	Mud Room: ceiling, 2'-9" from the right wall, 3'-5 1/2" from the rear wall	ceiling	33	
L24	4" recessed LED, 2700K, IC, dimmable	Mud Room: ceiling, 2'-9" from the right wall, 3'-7 1/2" from the front wall	ceiling	33	
R18	GFCI receptacle, 20A	Kitchen: on the left wall, 8'-1" from the rear wall	44"	11	counter, 4 ft spacing
R19	GFCI receptacle, 20A	Kitchen: on the left wall, 10'-6" from the rear wall	44"	11	counter, 4 ft spacing
R20	GFCI receptacle, 20A	Kitchen: on the left wall, 12'-1" from the front wall	44"	11	counter, 4 ft spacing
R21	GFCI receptacle, 20A	Kitchen: on the left wall, 8'-6" from the front wall	44"	11	counter, 4 ft spacing
R22	GFCI receptacle, 20A	Kitchen: on the front wall, 3'-9" from the left wall	44"	12	front counter
R23	GFCI receptacle, 20A	Kitchen: on the left wall, 12'-1" from the rear wall	27"	12	island end panel, living side, below the counter edge
R24	GFCI receptacle, 20A	Kitchen: on the left wall, 11'-6" from the front wall	27"	12	island, living side, below the counter edge
R25	240V receptacle or junction	Kitchen: on the left wall, 13'-6" from the rear wall	6"	1-2	range receptacle behind the range, NEMA 14-50R
L25	range hood, ducted outside	Kitchen: on the left wall, 13'-6" from the rear wall	72"	17	hood over the range, ducted outside
R26	receptacle, 20A dedicated	Kitchen: on the front wall, 5'-9 1/2" from the right wall	48"	14	refrigerator, above the floor in the fridge recess
R27	receptacle, 20A dedicated	Kitchen: ceiling, 5'-1/2" from the right wall, 9'-6" from the rear wall	12"	15	dishwasher, junction in the island sink base
R28	receptacle, 20A dedicated	Kitchen: ceiling, 5'-1/2" from the right wall, 11'-6" from the rear wall	12"	16	disposal, switched receptacle in the sink base; air switch on the sink deck
S10	dimmer, rocker	Kitchen: on the left wall, 2'-6" from the front wall	48"	34	3-gang beside the mud room opening: cans (3-way dimmer), pendants, under-cabinet; legs kit-cans, kit-pend, kit-ucl
S11	dimmer, rocker	Kitchen: on the left wall, 6'-1 1/2" from the rear wall	48"	34	3-way dimmer at the master hall opening; legs kit-cans
L26	4" recessed LED, 2700K, IC, dimmable	Kitchen: ceiling, 3'-5 1/2" from the left wall, 3'-6" from the rear wall	ceiling	34	
L27	4" recessed LED, 2700K, IC, dimmable	Kitchen: ceiling, 3'-5 1/2" from the left wall, 8'-6" from the rear wall	ceiling	34	
L28	4" recessed LED, 2700K, IC, dimmable	Kitchen: ceiling, 3'-5 1/2" from the left wall, 10'-3 1/2" from the front wall	ceiling	34	

Tag	Device	Position	AFF	Circuit	Notes
L29	4" recessed LED, 2700K, IC, dimmable	Kitchen: ceiling, 3'-5 1/2" from the left wall, 5'-1" from the front wall	ceiling	34	
L30	4" recessed LED, 2700K, IC, dimmable	Kitchen: ceiling, 1'-11" from the right wall, 3'-6" from the rear wall	ceiling	34	
L31	4" recessed LED, 2700K, IC, dimmable	Kitchen: ceiling, 1'-11" from the right wall, 8'-6" from the rear wall	ceiling	34	
L32	4" recessed LED, 2700K, IC, dimmable	Kitchen: ceiling, 1'-11" from the right wall, 8'-3 1/2" from the front wall	ceiling	34	
L33	4" recessed LED, 2700K, IC, dimmable	Kitchen: ceiling, 1'-11" from the right wall, 3'-3 1/2" from the front wall	ceiling	34	
L34	pendant	Kitchen: ceiling, 5'-1/2" from the right wall, 10'-6" from the rear wall	72"	34	over the island
L35	pendant	Kitchen: ceiling, 5'-1/2" from the right wall, 14'-3 1/2" from the front wall	72"	34	over the island
L36	pendant	Kitchen: ceiling, 5'-1/2" from the right wall, 10'-3 1/2" from the front wall	72"	34	over the island
L37	linear LED under cabinet	Kitchen: on the left wall, 9'-2" from the rear wall	54"	34	under the uppers, left of the range
L38	linear LED under cabinet	Kitchen: on the left wall, 10'-1/2" from the front wall	54"	34	under the uppers, right of the range
R29	GFCI receptacle, 20A	Dining: on the left wall, 2'-10 1/2" from the rear wall	16"	13	
R30	GFCI receptacle, 20A	Dining: on the left wall, 2'-6" from the front wall	16"	13	
R31	GFCI receptacle, 20A	Dining: on the rear wall, 4'-5" from the left wall	16"	13	between the windows
R32	GFCI receptacle, 20A	Dining: on the rear wall, 4'-7 1/2" from the right wall	16"	13	between the windows
R33	GFCI receptacle, 20A	Dining: on the right wall, 4'-10" from the rear wall	16"	13	between the windows
S12	dimmer, rocker	Dining: on the left wall, 9" from the front wall	48"	34	kitchen end of the dining wall; legs dining
L39	pendant	Dining: ceiling, 6'-5" from the right wall, 4'-6" from the front wall	78"	34	chandelier over the table; 30" over the top
R34	GFCI receptacle, 20A	Pantry: on the right wall, 4'-0" from the rear wall	44"	13	pantry counter
R35	GFCI receptacle, 20A	Pantry: on the left wall, 2'-2 1/2" from the rear wall	44"	13	pantry counter
S13	switch, rocker	Pantry: on the rear wall, 1'-10 1/2" from the right wall	48"	34	inside the opening; legs pantry
L40	surface LED, closet-rated (410.16)	Pantry: ceiling, 2'-9 1/2" from the left wall, 4'-0" from the rear wall	ceiling	34	
L41	surface LED, closet-rated (410.16)	Pantry: ceiling, 2'-8 1/2" from the right wall, 4'-0" from the rear wall	ceiling	34	
X1	media panel: Cat6 home runs, network switch	Pantry: on the left wall, 2'-1" from the front wall	54"	37	media panel on the mech room wall; every Cat6 home-runs here

Tag	Device	Position	AFF	Circuit	Notes
R36	boiler: 120V service switch + receptacle	Mechanical: on the right wall, 2'-5" from the rear wall	54"	31	heat source for the radiant slab (R-1)
R37	duplex receptacle, 15A	Mechanical: on the right wall, 1'-9 1/2" from the front wall	54"	31	circulators and zone control panel
R38	water heater: 120V receptacle (power vent)	Mechanical: on the right wall, 7" from the front wall	54"	32	
S14	switch, rocker	Mechanical: on the rear wall, 6" from the left wall	48"	34	inside the double doors; legs mech
L42	surface LED ceiling fixture	Mechanical: ceiling, 1'-6 1/2" from the left wall, 1'-8 1/2" from the rear wall	ceiling	34	
R39	duplex receptacle, 15A	Laundry: on the rear wall, 2'-2" from the right wall	24"	31	radiant manifold MF-1 actuator module (R-1)
R40	GFCI receptacle, 15A	Garage: on the rear wall, 10'-1 1/2" from the right wall	48"	31	radiant manifolds MF-2 / MF-G actuator module and the garage circulator P-RG (R-1)
R41	duplex receptacle, 15A	Bedroom Hall: on the left wall, 8'-5 1/2" from the rear wall	24"	31	radiant manifold MF-3 actuator module (R-1)
R42	duplex receptacle, 15A, tamper-resistant	Living: on the rear wall, 1'-2 1/2" from the left wall	16"	22	
R43	duplex receptacle, 15A, tamper-resistant	Living: on the rear wall, 6'-1" from the left wall	16"	22	
R44	duplex receptacle, 15A, tamper-resistant	Living: on the rear wall, 5'-11 1/2" from the right wall	16"	22	
R45	duplex receptacle, 15A, tamper-resistant	Living: on the rear wall, 1'-2" from the right wall	16"	22	
R46	duplex receptacle, 15A, tamper-resistant	Living: on the right wall, 1'-8" from the rear wall	16"	22	
R47	duplex receptacle, 15A, tamper-resistant	Living: on the right wall, 8'-9" from the rear wall	16"	22	
R48	duplex receptacle, 15A, tamper-resistant	Living: on the right wall, 7'-3 1/2" from the front wall	16"	22	
R49	duplex receptacle, 15A, tamper-resistant	Living: on the right wall, 1'-10 1/2" from the front wall	16"	22	
R50	recessed TV box: power + 2 x Cat6	Living: on the right wall, 12'-9" from the rear wall	66"	22	over the fireplace mantel
R51	duplex receptacle, 15A	Living: on the right wall, 10'-3 1/2" from the front wall	8"	22	fireplace insert blower, beside the insert
R52	duplex receptacle, 15A, tamper-resistant	Living: on the front wall, 8'-2" from the right wall	16"	22	
R53	duplex receptacle, 15A, tamper-resistant	Living: on the front wall, 2'-2" from the right wall	16"	22	

Tag	Device	Position	AFF	Circuit	Notes
R54	floor box receptacle	Living: ceiling, 8'-8 1/2" from the left wall, 8'-6" from the front wall	floor	22	floor box under the sofa (confirm the furniture layout)
S15	4-way switch, rocker	Living: on the rear wall, 6'-7" from the left wall	48"	34	4-gang beside the patio doors: living cans 4-way, living fan, patio lights, patio fan; legs living-cans, living-fan, patio-lights, patio-fan
S16	3-way switch, rocker	Living: on the right wall, 7'-5" from the rear wall	48"	34	at the bedroom hall opening; legs living-cans
L43	4" recessed LED, 2700K, IC, dimmable	Living: ceiling, 4'-1" from the left wall, 5'-6 1/2" from the rear wall	ceiling	34	
L44	4" recessed LED, 2700K, IC, dimmable	Living: ceiling, 4'-1" from the left wall, 5'-6" from the front wall	ceiling	34	
L45	4" recessed LED, 2700K, IC, dimmable	Living: ceiling, 10'-2" from the right wall, 5'-6 1/2" from the rear wall	ceiling	34	
L46	4" recessed LED, 2700K, IC, dimmable	Living: ceiling, 10'-2" from the right wall, 5'-6" from the front wall	ceiling	34	
L47	4" recessed LED, 2700K, IC, dimmable	Living: ceiling, 3'-11 1/2" from the right wall, 5'-6 1/2" from the rear wall	ceiling	34	
L48	4" recessed LED, 2700K, IC, dimmable	Living: ceiling, 3'-11 1/2" from the right wall, 5'-6" from the front wall	ceiling	34	
L49	fan-rated ceiling box, switched	Living: ceiling, 10'-2" from the right wall, 11'-1" from the front wall	ceiling	34	room centre, below the label
R55	duplex receptacle, 15A, tamper-resistant	Entry: on the left wall, 3'-8" from the front wall	16"	23	
S17	dimmer, rocker	Entry: on the right wall, 1'-2" from the front wall	48"	34	3-gang at the front door: living cans 3-way dimmer, entry light, porch lights; legs living-cans, entry, porch
L50	surface LED ceiling fixture	Entry: ceiling, 3'-6 1/2" from the right wall, 3'-10 1/2" from the front wall	ceiling	34	
A3	smoke + CO alarm, 120V interconnected w/ battery	Entry: ceiling, 2'-3 1/2" from the right wall, 2'-1 1/2" from the rear wall	ceiling	35	outside the office (a possible bedroom)
L51	surface LED, closet-rated (410.16), integral occupancy sensor	Coat Closet: ceiling, 1'-0" from the left wall, 1'-5 1/2" from the front wall	ceiling	34	
R56	duplex receptacle, 15A, tamper-resistant	Office: on the rear wall, 3'-2 1/2" from the left wall	16"	23	
R57	duplex receptacle, 15A, tamper-resistant	Office: on the rear wall, 3'-9" from the right wall	16"	23	
R58	duplex receptacle, 15A, tamper-resistant	Office: on the right wall, 4'-2 1/2" from the rear wall	16"	23	
R59	data outlet, 2 x Cat6	Office: on the right wall, 3'-11 1/2" from the front wall	16"	37	desk

Tag	Device	Position	AFF	Circuit	Notes
R60	duplex receptacle, 15A, tamper-resistant	Office: on the front wall, 2'-10" from the left wall	16"	23	
R61	duplex receptacle, 15A, tamper-resistant	Office: on the front wall, 1'-7" from the right wall	16"	23	
S18	dimmer, rocker	Office: on the left wall, 1'-8" from the front wall	48"	35	inside the French doors; legs office
L52	4" recessed LED, 2700K, IC, dimmable	Office: ceiling, 3'-2 1/2" from the left wall, 2'-10" from the rear wall	ceiling	35	
L53	4" recessed LED, 2700K, IC, dimmable	Office: ceiling, 3'-2 1/2" from the left wall, 2'-10 1/2" from the front wall	ceiling	35	
L54	4" recessed LED, 2700K, IC, dimmable	Office: ceiling, 3'-2" from the right wall, 2'-10" from the rear wall	ceiling	35	
L55	4" recessed LED, 2700K, IC, dimmable	Office: ceiling, 3'-2" from the right wall, 2'-10 1/2" from the front wall	ceiling	35	
A4	smoke alarm, 120V interconnected w/ battery	Office: ceiling, 6'-4 1/2" from the right wall, 2'-8" from the front wall	ceiling	35	
R62	duplex receptacle, 15A, tamper-resistant	Bedroom Hall: on the left wall, 9'-0" from the front wall	16"	24	hall over 10 ft (210.52(H))
S19	3-way switch, rocker	Bedroom Hall: on the left wall, 6'-0" from the rear wall	48"	35	living end, just past the opening; legs bhall
S20	3-way switch, rocker	Bedroom Hall: on the left wall, 1'-3 1/2" from the front wall	48"	35	bedroom 3 end; legs bhall
L56	4" recessed LED, 2700K, IC, dimmable	Bedroom Hall: ceiling, 2'-1/2" from the left wall, 3'-1/2" from the rear wall	ceiling	35	
L57	4" recessed LED, 2700K, IC, dimmable	Bedroom Hall: ceiling, 2'-1/2" from the left wall, 10'-0" from the front wall	ceiling	35	
L58	4" recessed LED, 2700K, IC, dimmable	Bedroom Hall: ceiling, 2'-1/2" from the left wall, 3'-7" from the front wall	ceiling	35	
A5	smoke + CO alarm, 120V interconnected w/ battery	Bedroom Hall: ceiling, 2'-1/2" from the left wall, 7'-5 1/2" from the rear wall	ceiling	35	outside the bedrooms
R63	duplex receptacle, 15A, tamper-resistant	Bedroom 1: on the rear wall, 5'-6" from the right wall	16"	24	between the windows
R64	duplex receptacle, 15A, tamper-resistant	Bedroom 1: on the front wall, 5'-1/2" from the left wall	16"	24	2 ft wall space between the door and the closet (210.52(A)(2))
R65	duplex receptacle, 15A, tamper-resistant	Bedroom 1: on the left wall, 3'-5 1/2" from the rear wall	16"	24	
R66	duplex receptacle, 15A, tamper-resistant	Bedroom 1: on the left wall, 3'-6 1/2" from the front wall	16"	24	

Tag	Device	Position	AFF	Circuit	Notes
R67	duplex receptacle, 15A, tamper-resistant	Bedroom 1: on the right wall, 6'-1/2" from the rear wall	16"	24	
R68	duplex receptacle, 15A, tamper-resistant	Bedroom 1: on the right wall, 2'-4" from the front wall	16"	24	
S21	switch, rocker	Bedroom 1: on the front wall, 4'-3 1/2" from the left wall	48"	35	2-gang, latch side of D17: light, fan; legs bed1-light, bed1-fan
L59	fan-rated ceiling box, switched	Bedroom 1: ceiling, 6'-6 1/2" from the left wall, 5'-8" from the rear wall	ceiling	35	
A6	smoke alarm, 120V interconnected w/ battery	Bedroom 1: ceiling, 2'-9 1/2" from the right wall, 3'-4" from the front wall	ceiling	35	
L60	surface LED, closet-rated (410.16), integral occupancy sensor	Bedroom 1 Closet: ceiling, 4'-4 1/2" from the left wall, 1'-0" from the rear wall	ceiling	35	
R69	GFCI receptacle, 20A	Bath 1: on the right wall, 3'-5" from the rear wall	40"	20	between the lav and the toilet
L61	wall sconce / vanity light	Bath 1: on the right wall, 1'-6 1/2" from the rear wall	84"	35	over the mirror
S22	switch, rocker	Bath 1: on the left wall, 3'-9" from the rear wall	48"	35	2-gang, latch side of D19: lights, fan; legs bath1-light, bath1-fan
L62	exhaust fan with integrated LED light, 110 CFM, humidity sensing	Bath 1: ceiling, 2'-1 1/2" from the left wall, 4'-8" from the rear wall	ceiling	35	
L63	4" recessed LED, wet location	Bath 1: ceiling, 2'-6 1/2" from the left wall, 1'-7 1/2" from the front wall	ceiling	35	in the shower
R70	duplex receptacle, 15A, tamper-resistant	Bedroom 2: on the rear wall, 4'-2" from the left wall	16"	25	
R71	duplex receptacle, 15A, tamper-resistant	Bedroom 2: on the right wall, 4'-8" from the rear wall	16"	25	between the windows
R72	duplex receptacle, 15A, tamper-resistant	Bedroom 2: on the front wall, 5'-4 1/2" from the left wall	16"	25	
R73	duplex receptacle, 15A, tamper-resistant	Bedroom 2: on the front wall, 3'-0" from the right wall	16"	25	
R74	duplex receptacle, 15A, tamper-resistant	Bedroom 2: on the left wall, 4'-1" from the rear wall	16"	25	
S23	switch, rocker	Bedroom 2: on the front wall, 7 1/2" from the left wall	48"	35	2-gang in the corner, latch side of D21: light, fan; legs bed2-light, bed2-fan
L64	fan-rated ceiling box, switched	Bedroom 2: ceiling, 7'-1" from the left wall, 5'-8" from the rear wall	ceiling	35	
A7	smoke alarm, 120V interconnected w/ battery	Bedroom 2: ceiling, 2'-9 1/2" from the right wall, 2'-6" from the front wall	ceiling	35	

Tag	Device	Position	AFF	Circuit	Notes
L65	surface LED, closet-rated (410.16), integral occupancy sensor	Closet (Bedroom 2): ceiling, 2'-6 1/2" from the left wall, 2'-3 1/2" from the rear wall	ceiling	35	
R75	GFCI receptacle, 20A	Bath 2: on the front wall, 4'-0" from the left wall	40"	20	between the two sinks
L66	wall sconce / vanity light	Bath 2: on the front wall, 2'-5 1/2" from the left wall	84"	35	over the mirror at sink 1
L67	wall sconce / vanity light	Bath 2: on the front wall, 5'-7" from the left wall	84"	35	over the mirror at sink 2
S24	switch, rocker	Bath 2: on the left wall, 2'-1 1/2" from the front wall	48"	35	2-gang, latch side of D22: lights, fan; legs bath2-light, bath2-fan
L68	exhaust fan with integrated LED light, 110 CFM, humidity sensing	Bath 2: ceiling, 6'-4 1/2" from the right wall, 2'-0" from the rear wall	ceiling	35	
L69	4" recessed LED, wet location	Bath 2: ceiling, 1'-4 1/2" from the right wall, 2'-9 1/2" from the front wall	ceiling	35	over the tub
R76	duplex receptacle, 15A, tamper-resistant	Bedroom 3: on the rear wall, 4'-11 1/2" from the right wall	16"	26	
R77	duplex receptacle, 15A, tamper-resistant	Bedroom 3: on the left wall, 5'-2 1/2" from the rear wall	16"	26	
R78	duplex receptacle, 15A, tamper-resistant	Bedroom 3: on the front wall, 2'-6 1/2" from the left wall	16"	26	
R79	duplex receptacle, 15A, tamper-resistant	Bedroom 3: on the front wall, 1'-11 1/2" from the right wall	16"	26	
R80	duplex receptacle, 15A, tamper-resistant	Bedroom 3: on the right wall, 4'-3 1/2" from the front wall	16"	26	
S25	switch, rocker	Bedroom 3: on the rear wall, 4'-1/2" from the left wall	48"	35	2-gang, latch side of D23: light, fan; legs bed3-light, bed3-fan
L70	fan-rated ceiling box, switched	Bedroom 3: ceiling, 6'-6" from the left wall, 5'-8" from the front wall	ceiling	35	
A8	smoke alarm, 120V interconnected w/ battery	Bedroom 3: ceiling, 2'-3" from the right wall, 2'-1" from the front wall	ceiling	35	
L71	surface LED, closet-rated (410.16), integral occupancy sensor	Closet (Bedroom 3): ceiling, 2'-7" from the right wall, 2'-5 1/2" from the front wall	ceiling	35	
X2	panel P1, 200A main breaker, 40 space	Garage: on the right wall, 10'-5 1/2" from the rear wall	60"	P1	beside the mud room door, 36" clear in front; fed from the meter-main outside
R81	GFCI receptacle, 20A	Garage: on the left wall, 17'-6 1/2" from the front wall	48"	28	between the garage doors
R82	GFCI receptacle, 20A	Garage: on the front wall, 4'-6 1/2" from the left wall	48"	28	
R83	GFCI receptacle, 20A	Garage: on the front wall, 7'-9" from the right wall	48"	28	
R84	GFCI receptacle, 20A	Garage: on the right wall, 10'-7 1/2" from the front wall	48"	28	

Tag	Device	Position	AFF	Circuit	Notes
R85	GFCI receptacle, 20A	Garage: on the rear wall, 10'-6 1/2" from the left wall	48"	28	
R86	GFCI receptacle, 20A	Garage: on the right wall, 5'-8" from the rear wall	48"	28	workbench
R87	240V receptacle, NEMA 14-50R	Garage: on the right wall, 3'-3" from the front wall	48"	9-10	EV-ready; swap for a hardwired charger if wanted
R88	GFCI receptacle, 20A	Garage: ceiling, 11'-1/2" from the left wall, 11'-6" from the rear wall	ceiling	29	door opener 1, ceiling
R89	GFCI receptacle, 20A	Garage: ceiling, 11'-1/2" from the left wall, 11'-0" from the front wall	ceiling	29	door opener 2, ceiling
S26	3-way switch, rocker	Garage: on the rear wall, 4'-8 1/2" from the left wall	48"	36	inside the service door, latch side of D3; legs garage
S27	3-way switch, rocker	Garage: on the rear wall, 7" from the right wall	48"	36	2-gang at the mud room door: garage lights 3-way, garage door lights; legs garage, garage-ext
L72	4 ft LED shop fixture	Garage: ceiling, 6'-1/2" from the left wall, 10'-10 1/2" from the rear wall	ceiling	36	
L73	4 ft LED shop fixture	Garage: ceiling, 6'-1/2" from the left wall, 9'-7 1/2" from the front wall	ceiling	36	
L74	4 ft LED shop fixture	Garage: ceiling, 12'-6 1/2" from the right wall, 10'-10 1/2" from the rear wall	ceiling	36	
L75	4 ft LED shop fixture	Garage: ceiling, 12'-6 1/2" from the right wall, 9'-7 1/2" from the front wall	ceiling	36	
L76	4 ft LED shop fixture	Garage: ceiling, 5'-1/2" from the right wall, 10'-10 1/2" from the rear wall	ceiling	36	
L77	4 ft LED shop fixture	Garage: ceiling, 5'-1/2" from the right wall, 9'-7 1/2" from the front wall	ceiling	36	
X3	meter-main: utility meter + 200A service disconnect (230.85)	Outside: outside, on the house wall, 31'-0" from the garage-door wall, 66'-5" from the master bedroom's rear wall	60"	P1	on the garage wall outside, beside the EV receptacle; utility service route to confirm
L78	4" recessed LED, wet location	Front Porch: porch/patio ceiling, 36'-0" from the garage-door wall, 58'-0" from the master bedroom's rear wall	ceiling	34	
L79	4" recessed LED, wet location	Front Porch: porch/patio ceiling, 47'-6 1/2" from the garage-door wall, 58'-0" from the master bedroom's rear wall	ceiling	34	
L80	4" recessed LED, wet location	Front Porch: porch/patio ceiling, 58'-0" from the garage-door wall, 58'-0" from the master bedroom's rear wall	ceiling	34	
L81	4" recessed LED, wet location	Front Porch: porch/patio ceiling, 70'-0" from the garage-door wall, 58'-0" from the master bedroom's rear wall	ceiling	34	

Tag	Device	Position	AFF	Circuit	Notes
L82	exterior LED, wet location, dark-sky	Front Porch: outside, on the house wall, 44'-11 1/2" from the garage-door wall, 53'-9" from the master bedroom's rear wall	72"	34	front door, left
L83	exterior LED, wet location, dark-sky	Front Porch: outside, on the house wall, 50'-2" from the garage-door wall, 53'-9" from the master bedroom's rear wall	72"	34	front door, right
R90	weatherproof GFCI receptacle, in-use cover	Front Porch: outside, on the house wall, 39'-9 1/2" from the garage-door wall, 53'-9" from the master bedroom's rear wall	24"	30	
R91	weatherproof GFCI receptacle, in-use cover	Front Porch: outside, on the house wall, 63'-0" from the garage-door wall, 53'-9" from the master bedroom's rear wall	24"	30	
L84	4" recessed LED, wet location	Covered Patio: porch/patio ceiling, 48'-6" from the garage-door wall, 5'-2 1/2" from the master bedroom's rear wall	ceiling	34	
L85	4" recessed LED, wet location	Covered Patio: porch/patio ceiling, 59'-8 1/2" from the garage-door wall, 5'-2 1/2" from the master bedroom's rear wall	ceiling	34	
L86	fan-rated ceiling box, wet location	Covered Patio: porch/patio ceiling, 54'-1" from the garage-door wall, 9'-0" from the master bedroom's rear wall	ceiling	34	
L87	exterior LED, wet location, dark-sky	Covered Patio: outside, on the house wall, 50'-4" from the garage-door wall, 16'-4 1/2" from the master bedroom's rear wall	72"	34	patio doors, left
L88	exterior LED, wet location, dark-sky	Covered Patio: outside, on the house wall, 57'-8" from the garage-door wall, 16'-4 1/2" from the master bedroom's rear wall	72"	34	patio doors, right
R92	weatherproof GFCI receptacle, in-use cover	Covered Patio: outside, on the house wall, 45'-2 1/2" from the garage-door wall, 16'-4 1/2" from the master bedroom's rear wall	24"	30	
R93	weatherproof GFCI receptacle, in-use cover	Covered Patio: outside, on the house wall, 63'-0" from the garage-door wall, 16'-4 1/2" from the master bedroom's rear wall	24"	30	
L89	exterior LED, wet location, dark-sky	Outside: outside, on the house wall, 5'-2" from the garage-door wall, 36'-8" from the master bedroom's rear wall	78"	36	service door
L90	exterior LED, wet location, dark-sky	Outside: outside, on the house wall, 0" from the garage-door wall, 39'-1" from the master bedroom's rear wall	84"	36	garage door coach light
L91	exterior LED, wet location, dark-sky	Outside: outside, on the house wall, 0" from the garage-door wall, 52'-1" from the master bedroom's rear wall	84"	36	garage door coach light

Tag	Device	Position	AFF	Circuit	Notes
L92	exterior LED, wet location, dark-sky	Outside: outside, on the house wall, 0" from the garage-door wall, 65'-1" from the master bedroom's rear wall	84"	36	garage door coach light
X4	fused disconnect DS1, CU-1	Outside: outside, on the house wall, 5'-4 1/2" from the garage-door wall, 31'-7" from the master bedroom's rear wall	48"	5-6	left side yard, on the master closet wall beside the unit (M-1)
R94	weatherproof GFCI receptacle, in-use cover	Outside: outside, on the house wall, 5'-4 1/2" from the garage-door wall, 25'-7" from the master bedroom's rear wall	24"	30	service receptacle within 25 ft of CU-1 (210.63)
X5	fused disconnect DS2, CU-2	Outside: outside, on the house wall, 83'-5 1/2" from the garage-door wall, 45'-0" from the master bedroom's rear wall	48"	7-8	right side yard, on the bedroom 3 closet wall beside the unit (M-1)
R95	weatherproof GFCI receptacle, in-use cover	Outside: outside, on the house wall, 83'-5 1/2" from the garage-door wall, 40'-11" from the master bedroom's rear wall	24"	30	service receptacle within 25 ft of CU-2 (210.63)
X6	low-voltage drop: Cat6 for a video doorbell	Outside: outside, on the house wall, 50'-11" from the garage-door wall, 53'-9" from the master bedroom's rear wall	48"	37	right of the front door

Service load calculation (NEC 220.82)

Item	Basis	Load
General lighting and receptacles	3 VA x 3,185 sf	9,555 VA
Small-appliance circuits	3 × 1,500 VA (11, 12, 13)	4,500 VA
Laundry circuit	1 × 1,500 VA (18)	1,500 VA
Fixed appliances, nameplate	range, dryer, EV-ready, fridge, DW, disposal, hood, openers, boiler, water heater, media	32,500 VA
All other load, demand	first 10 kVA at 100%, rest at 40% (220.82(B))	25,222 VA
Heating / cooling	two mini-split outdoor units at 100% (220.82(C)); radiant boiler is gas	10,800 VA
Calculated load	36,022 VA / 240 V	150 A on a 200 A service

Bill of materials

Rough-order quantities from the device layout, for pricing. The electrician takes off the final list.

Item	Qty	Notes
Duplex receptacles, TR	45	15A, with boxes
Duplex + USB-C receptacles	2	nightstands
GFCI receptacles, 20A	29	kitchen, dining, baths, laundry, pantry, garage
Weatherproof GFCI receptacles	6	in-use covers

Item	Qty	Notes
Dedicated / 240V receptacles	9	washer, fridge, DW, disposal, boiler, water heater, range 14-50, dryer 14-30, EV 14-50
Floor box	1	living
Switches and dimmers	27	single-pole, 3-way, 4-way, dimmers, one occupancy sensor; gangs per the switch plan
Recessed LED cans	42	4", 2700K; sloped trims in the kitchen and living, wet-rated in showers, tubs, porch and patio
Surface / closet fixtures	18	closet-rated with occupancy sensors in the small closets; 4 ft LED shop fixtures in the garage
Pendants and chandelier	4	island and dining
Vanity lights and exterior lights	14	exterior dark-sky
Fan-rated boxes	6	master, living, three bedrooms, patio
Bath exhaust fans	5	110 CFM, humidity sensing, integrated light
Smoke and smoke + CO alarms	8	hardwired, interconnected
Panel and meter-main	1 + 1	200A, 40 spaces, AFCI/GFCI breakers per the schedule; meter-main with the 230.85 disconnect
Fused disconnects	2	one per mini-split outdoor unit
Media panel, Cat6	1	4 drops, ~220 ft of Cat6 (2 per TV box and desk)
NM-B 14/2, 14/3, 12/2, 10/3, 6/3	~5,500 ft	rough order from the device count (about 24 ft per device)